Planogram Compliance and Shelf Monitoring (Public Research Summary)

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1 Source and licence

Frontoni, E., Mancini, A. and Zingaretti, P. (2015). Embedded Vision Sensor Network for Planogram Maintenance in Retail Environments. *Sensors*, 15(9), 21114-21133. https://doi.org/10.3390/s150921114 (open access, PubMed Central PMC4610585).

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2 What a planogram is

The authors describe a planogram as a detailed visual map of where each product should be placed in a store. Retailers and manufacturers use planograms to control which products get space, how many facings each gets, and where they sit on the shelf, with the aim of increasing sales and profit. If the shelf does not follow the planogram, those aims are undermined.

3 Why compliance matters

- The article cites a study by the National Association for Retailing Merchandising Services (NARMS) reporting that **100% planogram compliance** was associated with a **7.8% sales lift** and an **8.1% profit improvement**.
- The authors argue that keeping the shelf consistent with the planogram is key to increasing shelf value and improving sell-out.
- Compliance information also lets brands and retailers compare stores, and can support premiums or penalties in agreements between them.

4 Why shelves drift from the planogram

The article's trial in real stores observed several reasons why the actual shelf differs from the plan, or why it is hard to check:

- Store staff and refill teams placing products in the wrong position.
- Products being moved, rotated or partially hidden.
- Out-of-stock gaps, which can look similar to a product that has been moved.
- Practical conditions such as shoppers passing in front of the shelf and changing light.

5 How compliance is checked

The authors note that commercial planogram tools of the time relied on a person taking photos in front of the shelf, which were analysed afterwards on a web or desktop platform. These checks are periodic, not real time. The article proposes a network of low-cost cameras to monitor shelves continuously and detect planogram errors automatically.

6 What this means for Marvel's field teams

- A field audit is a snapshot. Planogram compliance can change between visits, so always check the audit date before relying on it (MCPG-06).
- Planogram errors and stock-outs are linked: a gap on the shelf can be either, and both reduce what shoppers can find.
- Use the NARMS figure as **industry context only**. It comes from another study cited in this article, not from Marvel's data. Marvel's own data shows no measurable sales difference from planogram compliance alone, so representatives must not promise the 7.8% figure to an account (MCPG-03, section 7; MCPG-04, section 2.9).

7 Limitations

- The NARMS figures are reported second-hand in this article; the original NARMS study is not reviewed here.
- The article's main contribution is a monitoring technology, tested in a limited number of real store settings.

Document control

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Open points and review notes

- Summary written by Team 1; key statements checked against the article text on 20 October 2026.